

Overview

This guide covers the Nifty 100 analytics workflow, source data, and reproducible outputs.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Setup

Create the virtual environment, install requirements, run `python -m src.etl.database`, then start the dashboard and API.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Dashboard Home

Use the year selector to inspect KPI summaries, sector composition, and quality-ranked companies.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Screener

Adjust threshold controls and download the visible result table as CSV. Missing metrics remain N/A.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Company Profiles

Search a ticker to inspect annual history, profitability, leverage, cash flow, and generated reports.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Reports

Run `python -m src.reports.tearsheet` for company PDFs, `sector_report` for sector summaries, and `portfolio_report` for the portfolio PDF.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Clustering

Run `python -m src.analytics.clustering` to create five reproducible financial archetypes, the elbow plot, portfolio statistics, and outlier report.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

API

Start uvicorn src.api.main:app --port 8000. Open /docs for Swagger. Example: curl <http://localhost:8000/api/v1/companies/TCS/ratios>.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Troubleshooting

Rebuild nifty100.db with `python -m src.etl.database` if tables are missing. Partial source histories are expected and are reported rather than fabricated.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.

Data caveats

The supplied source has 101 companies rather than the planning target of 92. AGTL is skipped from tearsheets because it has fewer than three P&L; years.

Operational example and expected behavior

Use the generated CSV, workbook, or API JSON as the audit trail for this workflow. All generated values retain missing-data semantics and are suitable for review.